

Exercise 1 A given Hilbert space is spanned by n -kets $|\alpha_i\rangle$ with $i = 1, \dots, n$. In this space I define operators S_i that act on the basis vectors in the following way:

$$S_i|\alpha_j\rangle = [c_j|\alpha_i\rangle + c_i|\alpha_j\rangle]|\varepsilon_{ij}| + \delta_{ij}|\alpha_j\rangle \quad (1)$$

where ε_{ij} is an antisymmetric tensor satisfying $\varepsilon_{ij} = -\varepsilon_{ji} - 1$, ε_{ij} denote the absolute value, δ_{ij} is the Kronecker delta function and c_i are some non-zero real constants that satisfy the constraint:

$$\sum_{i=1}^n c_i = -(n-1). \quad (0.2) \quad (2)$$

Determine the matrix representation of an operator S_k where $1 \leq k \leq n$.

The matrix element $(S_i)_{mj}$ is given by:

$$\langle\alpha_m|S_i|\alpha_j\rangle = (c_j\delta_{mi} + c_i\delta_{mj})|\varepsilon_{ij}| + \delta_{ij}\delta_{mj}$$

We consider the case $i = j$. Then, the antisymmetric metric tensor $\varepsilon_{ii} = -e_{ii} = 0$ and $\delta_{ii=1}$ so we get

$$\langle\alpha_m|S_i|\alpha_j\rangle = \delta_{ij}\delta_{mj}$$

If $i \neq j$, then $\delta_{ij} = 0$ and $|\varepsilon_{ij}| = 1$, so

$$\langle\alpha_m|S_i|\alpha_j\rangle = (c_j\delta_{mi} + c_i\delta_{mj})$$

Let's consider the diagonal elements $(S_i)_{mm}$. If $i = m$, then we look at the first case and see that $(S_i)_{ii} = 1$. Else, we see that $(S_i)_{mm} = (c_m\delta_{mi} + c_i\delta_{mm})|\varepsilon_{im}| = c_i$. Any off diagonal terms occur at $m \neq j$ so that $\delta_{ij}\delta_{mj} = 0$. Hence, we look at the second case and see that $(S_i)_{mj} = c_j\delta_{mi}$. So off diagonal terms occur along row $m = i$ with value c_j as indexed by the column. Hence,

$$S_1 = \begin{pmatrix} 1 & c_2 & c_3 & \dots & c_n \\ 0 & c_1 & 0 & \dots & 0 \\ 0 & 0 & c_1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & c_1 \end{pmatrix} \quad S_i = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & \ddots & 0 & \dots & 0 \\ c_1 & c_2 \dots c_{i-1} & 1 & c_{i+1} \dots & c_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & c_i \end{pmatrix}$$

Some c_j has to be non-zero by the constraint condition. We then see that $(S_i)_{ij} = c_j$ but $(S_i)_{ji} = 0 \neq c_j^\dagger$ for $i \neq j$. Thus, S_i is **not Hermitian** for any i . The trace of S_i is the sum of the diagonal entries, which gives $(n-1)c_i + 1$. Hence, $\text{tr}(S_i) = (n-1)c_i + 1$. Then,

$$\text{tr}(S) = \sum_{i=1}^n \text{tr}(S_i) = (n-1) \sum_{i=1}^n c_i + n = -n + n = 0$$

We also note that S is **not Hermitian** since $S_{k1} = c_1$ but $S_{1k} = c_k$.

Exercise 2a In a Stern-Gerlach experiment I send a beam of silver atoms through a magnetic field that is pointing strictly along the z -direction i.e ignoring the $\nabla \cdot \vec{B} = 0$ constraint. The beam would naturally split into $\pm z$ directions. Now if I want to measure J_x component of the spin, do I detect any non-zero result? Give reason for your answer.

Yes. The two beams correspond to the $|\pm z\rangle$ eigenstates which can collapse into the $|\pm x\rangle$ with equal probability when J_x is measured.

Exercise 2b Imagine due to some reason the Planck's constant $\hbar$ doubles its value i.e $\hbar_{new} = 2\hbar$. This would mean, in terms of the original $\hbar$, all particles would effectively behave as integer spin particles. What would you expect to see in a Stern-Gerlach experiment if I now send silver atoms through magnetic field oriented along z direction? Give reason for your answer.

The electron now behaves as a spin-1 particle so that eigenstates of S_z are available. These will quantize the magnetic dipole moment $\mu \in [-1, 0, 1]\hbar$ so that 3 peaks will be seen on the screen. One corresponding to 0 deflection due to the $m_s = 0$ eigenstate and two peaks symmetric about the 0 deflection line corresponding to $m_s = \pm\hbar$.

Exercise 2c I construct a Stern-Gerlach type experiment but with the following changes: (i) I replace the silver atoms by electrons, and I replace the magnetic field by a strong electric field, (ii) I replace the silver atoms by electrons, but I keep the magnetic field the same, (iii) I keep the silver atoms, but I replace the magnetic field by a strong electric field, and (iv) I keep the silver atoms, but I make the magnetic field constant along z -direction. Do I expect to see any effects on the screen? Give only qualitative answers to the questions

- (i) The magnetic dipole moment does not interact with the electric field so the spin does not contribute. However, the electrons experience a force in the opposite direction to the electric field due to the coulomb force. So one peak emerges on the screen due to this force, which is the same for each electron.
- (ii) The same deflection pattern is observed as in the original experiment as the electrons have the same magnetic moment due to intrinsic spin.
- (iii) The magnetic moment does not interact with the electric field and since the silver atoms are electrically neutral, no deflection is seen at all.
- (iv) No deflection is seen as the force on the atom is given by $F_z = -\partial_z(\vec{\mu} \cdot \vec{B}) = \mu \partial_{B_z} z$, so if B is constant in the z direction, the force goes to 0.

Exercise 2d An electron is known to have zero size, i.e it is exactly point like. Imagine somebody tells you that an electron is not quite point like but it is more like a small ring of thickness δ and inner radius r , where $(r, \delta) \rightarrow 0$ of course. Would you be able to refute or agree with his statement using only Stern-Gerlach experiment with silver atoms? Give qualitative reasonings. Assume that the nucleus remains as before and only the electrons change in the aforementioned way.

You would expect each electron to have a classical angular momentum being a spinning ring. The results of the stern-gerlach experiment would necessitate that this angular momentum was quantized. This would imply that each electron ring spins at the same speed and the quantization is due to it spinning in one direction or another.

Exercise 3a If an operator A commutes with both J_+ and J_- , show that it also commutes with J_z , i.e $[A, J_z] = 0$. Here $J_{\pm}$ are the raising-lowering operators. Does A commute with J^2 , the total angular momentum?

Note that

$$J_+J_- = (J_x + iJ_y)(J_x - iJ_y) = J_x^2 - i[J_x, J_y] + J_y^2 = J_x^2 + J_y^2 + \hbar J_z$$

Hence,

$$J_z = \frac{1}{\hbar}(J_+J_- - J_x^2 - J_y^2)$$

A necessarily commutes with J_x and J_y since these operators are formed from linear combinations of $J_{\pm}$. Thus, A commutes with J_z since it commutes with $J_{\pm}$ and J_x, J_y . Since A commutes with each of J_x, J_y, J_z , it commutes with J^2 .

Exercise 3c In a certain n -dimensional Hilbert space I have a traceless operator A that has only two distinct eigenvalues each with certain number of degeneracies. Show that this information can at least specify one of the eigenvalue, say λ , in terms of $\det A$ as $\lambda \propto (\det A)^{1/n}$

Proof. Let A have eigenvalues λ and κ . If λ has degeneracy p , then κ necessarily has degeneracy $n - p$. The traceless condition gives us that

$$p\lambda + (n - p)\kappa = 0 \implies \kappa = \frac{p}{p - n}\lambda$$

The determinant of A is given by the product of eigenvalues (with multiplicity) so that

$$\det A = \lambda^p \kappa^{n-p} = \lambda^p \left(\frac{p}{p - n} \lambda \right)^{n-p} = \lambda^n \left(\frac{p}{p - n} \right)^{n-p}$$

Thus,

$$\lambda = \left(\frac{p}{p - n} \right)^{(n-p)/n} (\det A)^{1/n} \propto (\det A)^{1/n}$$

as required. □

Exercise 3d In a certain n -dimensional Hilbert space, an operator A has n eigenvalues λ_i with $i = 1, \dots, n$ such that $\lambda_i \leq \lambda_{i+1}$. Argue that in this space expectation value of A in any generic state $|\psi\rangle$ satisfy $\lambda_1 \leq \langle \psi | A | \psi \rangle$ assuming the normalisability condition $\langle \psi | \psi \rangle = 1$.

Let $|i\rangle$ be the normalized eigenstate corresponding to eigenvalue λ_i so that $\{|i\rangle\}_{i=1,\dots,n}$

is an orthonormal basis. Expand $|\psi\rangle = \sum_m c_m |m\rangle$. Then,

$$\langle\psi|A|\psi\rangle = \sum_{m,k} \langle m|c_m^* A c_k|k\rangle = \sum_{m,k} c_m^* c_k \lambda_k \langle m|k\rangle = \sum_m \lambda_m |c_m|^2$$

Normalization of ψ dictates that $\sum_m |c_m|^2 = 1$. We then see that

$$\begin{aligned}\langle\psi|A|\psi\rangle &\geq \sum_m \lambda_1 |c_m|^2 = \lambda_1 \sum_m |c_m|^2 = \lambda_1 \\ \langle\psi|A|\psi\rangle &\leq \sum_m \lambda_n |c_m|^2 = \lambda_n \sum_m |c_m|^2 = \lambda_n\end{aligned}$$

where the inequalities hold because $\lambda_1 \leq \lambda_i \leq \lambda_n$ by assumption.

Exercise 4 Using the commutator relation $[J_x, J_y] = i\hbar J_z$ and cyclic permutations between them, show that:

$$J_-|j, m\rangle = \hbar\sqrt{j(j+1) - m(m-1)}|j, m-1\rangle$$

where j is the highest eigenvalue of J^2 in a given Hilbert space and J_- is the lowering operator. For a spin $1/2$ particle with spin pointing along z direction, why isn't the total angular momentum J^2 simply given by $\hbar^2/4$?

Proof. First we show that $J_-|j, m\rangle$ is an eigenstate of J_z . Observe that

$$\begin{aligned} [J_z, J_-] &= J_z(J_x - iJ_y) - (J_x - iJ_y)J_z \\ &= [J_z, J_x] + i[J_y, J_z] \\ &= i\hbar J_y - \hbar J_x = -\hbar J_- \end{aligned}$$

Now we see that

$$\begin{aligned} J_z(J_-|j, m\rangle) &= ([J_z, J_-] + J_-J_z)|j, m\rangle \\ &= -\hbar J_-|j, m\rangle + m\hbar J_-|j, m\rangle \\ &= (m-1)\hbar J_-|j, m\rangle \end{aligned}$$

so that $J_-|j, m\rangle$ is an eigenstate of J_z with eigenvalue $(m-1)\hbar$. We claim that $J_-|j, m\rangle = \alpha|j, m-1\rangle$. We know that there can be no contributions from eigenkets with other m values as they produce orthogonal eigenstates. Further, $[J_-, J^2] = [J_x - iJ_y, J^2] = 0$ as each J_i commutes with J^2 . Thus, $J_-|j, m\rangle \neq \alpha|j', m-1\rangle$ as this does not commute with J^2 . Now we see that

$$\begin{aligned} \alpha^2 &= \langle j, m | J_-^\dagger J_- | j, m \rangle \\ &= \langle j, m | ((J_x + iJ_y)(J_x - iJ_y)) | j, m \rangle \\ &= \langle j, m | J_x^2 + J_y^2 - i[J_x, J_y] | j, m \rangle \\ &= \langle j, m | J^2 - J_z^2 + \hbar J_z | j, m \rangle \\ &= j(j+1)\hbar^2 - m^2\hbar^2 + m\hbar^2 \\ &= \hbar^2[j(j+1) - m(m-1)] \end{aligned}$$

Thus,

$$\alpha = \hbar\sqrt{j(j+1) - m(m-1)}$$

as required.

The total angular momentum of a spin $1/2$ system ($j = 1/2$) prepared in the $m = 1/2$ state is $J^2|1/2, 1/2\rangle = 1/2(1/2 + 1)\hbar^2 = 3\hbar^2/4$. This is because the state can still have S_x and S_y components as $S_{x/y}j, m \neq 0$. $\square$

Exercise 5a In a two-dimensional Hilbert space spanned by eigenstates of J_z an electron is in $\langle +z \rangle$ state. With what uncertainties can I measure J_x and J_y angular momenta? Determine the lower bound satisfied by the product of these uncertainties. Do not derive the relation.

We have that

$$\begin{aligned}\langle J_x \rangle &= \langle +z | J_x | +z \rangle = \frac{\hbar}{2} \langle +z | -z \rangle = 0 \\ \langle J_x^2 \rangle &= \langle +z | J_x | +z \rangle = \frac{\hbar}{2} \langle +z | J_x | -z \rangle = \frac{\hbar^2}{4}\end{aligned}$$

Thus, the uncertainty with which you can measure J_x is

$$\Delta J_x = \sqrt{\langle J_x^2 \rangle - \langle J_x \rangle^2} = \frac{\hbar}{2}$$

Similarly,

$$\begin{aligned}\langle J_y \rangle &= \langle +z | J_y | +z \rangle = \frac{i\hbar}{2} \langle +z | -z \rangle = 0 \\ \langle J_y^2 \rangle &= \langle +z | J_y | +z \rangle = i\frac{\hbar}{2} \langle +z | J_y | -z \rangle = -i^2 \frac{\hbar^2}{4} = \frac{\hbar^2}{4}\end{aligned}$$

So

$$\Delta J_y = \sqrt{\langle J_y^2 \rangle - \langle J_y \rangle^2} = \frac{\hbar}{2}$$

The uncertainty relation gives us a lower bound on the product of these uncertainties:

$$\Delta J_x \Delta J_y \geq \frac{1}{2} |\langle J_x, J_y \rangle| = \frac{1}{2} |i\hbar \langle J_z \rangle| = \frac{1}{2} |i\hbar \frac{\hbar}{2}| = \frac{\hbar^2}{4}$$

Exercise 5b The raising (or lowering) operator satisfy the commutator relation $[J_z, J_+] = \hbar J_+$ with J_z . Using this can we say that the product of the uncertainties in measuring J_z and J_+ has a lower bound of $\frac{\hbar}{2} |\langle J_+ \rangle|$? Would there be any apparent contradiction with known results?

No we cannot conclude that lower bound because J_+ is not hermitian and the proof of the bound requires hermiticity. This is also clearly violated for the state $|+z\rangle$ which is an eigenstate of J_z and J_+ with eigenvalues $\hbar/2$ and 0 respectively. In this case, $\Delta J_z = \Delta J_+ = 0$.

Exercise 5c An operator A commutes with another operator C in a given Hilbert space. This means that we can measure them simultaneously and if the basis of the Hilbert space are eigenstates of C , then there are no uncertainties in the measurements

of A and C . However the operator could also be thought of as being made of operators A_i such that

$$A = \sum_{i=1}^n A_i \quad (3)$$

where n is some integer. Will there be uncertainties in C and A_i now? Give an example to justify your result.

There can be. Consider $J^2 = J_x^2 + J_y^2 + J_z^2$. J^2 commutes with J_z , but J_z does not commute with J_x^2 or J_y^2 so there can be uncertainties in these measurements.

Exercise 5d Two operators A and B anti-commute to a third operator C in a given Hilbert space:

$$\{A, B\} = AB + BA = C \quad (4)$$

In this space however there are uncertainties in the measurement of both A and B with the following lower bound: $\Delta A \Delta B \geq 3\hbar$. With these informations, can you determine the operator C ?

I don't think so.

Exercise 6a In a given n -dimensional Hilbert space a Hermitian operator A has n non-degenerate eigenvalues λ_k where $k = 1, \dots, n$. Show that a generic operator like A can alternatively be written as:

$$A = \sum_{k=1}^n \lambda_k |\psi_k\rangle\langle\psi_k|$$

where $|\psi_k\rangle$ form the basis vectors of the Hilbert space with $k = 1, \dots, n$ as before.

The orthonormality of the basis gives

$$\sum_{k=1}^n |\psi_k\rangle\langle\psi_k| = 1$$

Thus,

$$A = A \sum_{k=1}^n |\psi_k\rangle\langle\psi_k| = \sum_{k=1}^n A |\psi_k\rangle\langle\psi_k| = \sum_{k=1}^n \lambda_k |\psi_k\rangle\langle\psi_k|$$

as desired.

Exercise 6b In a two-dimensional Hilbert space an operator has two eigenstates given by

$$|\psi_1\rangle = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}, \quad |\psi_2\rangle = \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} \quad (5)$$

corresponding to the two eigenvalues λ_1 and λ_2 respectively with $\lambda_1 \neq \lambda_2$. Using this information only, determine the matrix representation of the operator A . Assume that the (a_i, b_i, λ_i) with $i = 1, 2$ are given integers.

Let A_ψ denote the operator A represented in the basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ and A_0 the operator represented in the given basis. We know that A_ψ is diagonal with entries λ_i . Note that

$$P = \begin{pmatrix} \uparrow & \uparrow \\ |\psi_1\rangle & |\psi_2\rangle \\ \downarrow & \downarrow \end{pmatrix} = \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}$$

is the change of basis matrix between the basis $|\psi_i\rangle$ basis and the given one. Thus,

$$\begin{aligned}
 A_0 &= PA_\psi P^{-1} \\
 &= \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \frac{1}{a_1 b_2 - a_2 b_1} \begin{pmatrix} b_2 & -a_2 \\ -b_1 & a_1 \end{pmatrix} \\
 &= \frac{1}{a_1 b_2 - a_2 b_1} \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix} \begin{pmatrix} \lambda_1 b_2 & -\lambda_1 a_2 \\ -\lambda_2 b_1 & \lambda_2 a_1 \end{pmatrix} \\
 &= \frac{1}{a_1 b_2 - a_2 b_1} \begin{pmatrix} a_1 b_2 \lambda_1 - a_2 b_1 \lambda_2 & -a_1 a_2 \lambda_1 + a_1 a_2 \lambda_2 \\ b_1 b_2 \lambda_1 - b_1 b_2 \lambda_2 & -b_1 a_2 \lambda_1 + a_1 b_2 \lambda_2 \end{pmatrix} \\
 &= \frac{1}{a_1 b_2 - a_2 b_1} \begin{pmatrix} a_1 b_2 \lambda_1 - a_2 b_1 \lambda_2 & -a_1 a_2 (\lambda_1 - \lambda_2) \\ b_1 b_2 (\lambda_1 - \lambda_2) & -a_2 b_1 \lambda_1 + a_1 b_2 \lambda_2 \end{pmatrix}
 \end{aligned}$$

Exercise 6c Show that if two operators A and B commute and $|\psi_1\rangle$ is an eigenstate of A with an eigenvalue λ , then the state $|\psi_2\rangle = B|\psi_1\rangle$ is also an eigenstate of A with the same eigenvalue. On the other hand show that if there exists a basis of eigenvectors common to both the operators A and B then these two operators must commute.

Proof. If A and B commute, then $AB = BA$ and we see that

$$A|\psi_2\rangle = AB|\psi_1\rangle = BA|\psi_1\rangle = \lambda B|\psi_1\rangle = \lambda|\psi_2\rangle$$

So $|\psi_2\rangle$ is eigenstate of A with same eigenvalue.

Let $\{|m_i\rangle\}$ be a basis of eigenstates with eigenvalues a_i and b_i for A and B respectively. First note that for an arbitrary basis vector $|m_i\rangle$, we have

$$AB|m_i\rangle = b_i A|m_i\rangle = a_i b_i |m_i\rangle = BA|m_i\rangle$$

Then, for an arbitrary ket $|\psi\rangle = \sum_i c_i |m_i\rangle$, we have

$$AB|\psi\rangle = \sum_i c_i AB|m_i\rangle = \sum_i c_i BA|m_i\rangle = BA|\psi\rangle$$

So A and B commute. □

Exercise 6d I define the so called parity operator Π satisfying the standard operator equation

$$\Pi|x\rangle = |-x\rangle \quad (6)$$

where x is the coordinate along x -direction. Using this definition determine the action of the following

$$\sin\left[\alpha\left(\frac{1+\Pi}{2}\right)\right]|x\rangle \quad (7)$$

where α is a positive integer. Observe that for our case there is no constraint on α .

Note that $(1 + \Pi)|x\rangle = |x\rangle + |-x\rangle$ and $(1 + \Pi)(|x\rangle + |-x\rangle) = |x\rangle + |-x\rangle + |-x\rangle + |x\rangle = 2(|x\rangle + |-x\rangle)$. Thus, it follows that

$$(1 + \Pi)^n |x\rangle = 2^{n-1}(|x\rangle + |-x\rangle), \quad n \geq 1$$

Not worrying about convergence for now, we take the sin of the operator to be

$$\begin{aligned} \sin \left[\alpha \left(\frac{1 + \Pi}{2} \right) \right] |x\rangle &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\alpha^{2n-1} (1 + \Pi)^{2n-1}}{2^{2n-1} n!} |x\rangle \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{\alpha^{2n-1}}{2^{2n-1} n!} 2^{2n-2} (|x\rangle + |-x\rangle) \\ &= \left(\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\alpha^{2n-1}}{n!} \right) \frac{1}{2} (|x\rangle + |-x\rangle) \\ &= \frac{1}{2} \sin(\alpha) (|x\rangle + |-x\rangle) \end{aligned}$$

Exercise 7 A two dimensional Hilbert space is spanned by states: $|\phi_1\rangle$ and $|\phi_2\rangle$ that are eigenvectors of an operator H_0 with eigenvalues E_1 and E_2 . Imagine that I add a small shift to the operator H_0 via a matrix W :

$$W = \begin{pmatrix} W_{11} & W_{12} \\ W_{12}^* & W_{22} \end{pmatrix} \quad (8)$$

with W_{11} and W_{22} real, such that the shifted operator is $H = H_0 + W$. Clearly for this shift, the states $|\phi_1\rangle$ and $|\phi_2\rangle$ are no longer eigenstates of the new operator H . Let us denote the new eigenstates by $|\psi_{\pm}\rangle$. Determine these new eigenstates and the new eigenvalues of H . Show that you can always express these states as:

$$\begin{pmatrix} |\psi_+\rangle \\ |\psi_-\rangle \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} & \sin \frac{\theta}{2} e^{i\varphi/2} \\ -\sin \frac{\theta}{2} e^{-i\varphi/2} & \cos \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix} \begin{pmatrix} |\phi_1\rangle \\ |\phi_2\rangle \end{pmatrix} \quad (9)$$

where the angles θ ($0 \leq \theta \leq \pi$) and φ are defined as:

$$\tan \theta = \frac{2|W_{12}|}{E_1 + W_{11} - E_2 - W_{22}}, \quad e^{i\varphi} = \frac{W_{21}}{|W_{21}|} \quad (10)$$

We are looking at the matrix

$$H = \begin{pmatrix} E_1 + W_{11} & W_{12} \\ W_{12}^* & E_2 + W_{22} \end{pmatrix}$$

Let $\alpha = E_1 + W_{11} - E_2 - W_{22}$. Mathematica yields eigenvalues

$$E_{\pm} = \frac{1}{2}(E_1 + W_{11} + E_2 + W_{22} \pm \sqrt{\alpha^2 + 4|W_{21}|^2})$$

and eigenvectors

$$|\psi_{\pm}\rangle = (\alpha \mp \sqrt{\alpha^2 + 4|W_{21}|^2}, 2W_{21})$$

Letting $\sim$ denote equal up to a multiplicative factor, we find

$$\begin{aligned} |\psi_-\rangle &\sim \left(1 - \sqrt{1 + \frac{4|W_{21}|^2}{\alpha^2}}, \frac{2W_{21}}{\alpha} \right) \\ &\sim (1 - \sqrt{1 + \tan^2 \theta}, \tan \theta e^{i\varphi}) \\ &\sim (\cos \theta - 1, \sin \theta e^{i\varphi}) \end{aligned}$$

Now we normalize by dividing by $|(\cos \theta - 1, \sin \theta e^{i\varphi})|^2 = \sqrt{2 - 2 \cos \theta} =$

$\sqrt{2(1 - (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}))} = \sqrt{4 \sin^2 \frac{\theta}{2}} = 2 \sin \frac{\theta}{2}$. Then, we get

$$\begin{aligned} |\psi_{-}\rangle &\sim \left(\frac{\cos \theta - 1}{2 \sin \frac{\theta}{2}}, \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} e^{i\varphi} \right) \\ &\sim \left(-\sin \frac{\theta}{2}, \cos \frac{\theta}{2} e^{i\varphi} \right) \end{aligned}$$

splitting the $e^{i\varphi}$ between the two components gives the desired result. The same can be done for $|\psi_{+}\rangle$.

Exercise 8a A linear array of n atoms is arranged so that electrons can hop from one atom to another. Each of these atoms represent a state in a Hilbert space. Determine the matrix representation of a shift operator that, when acting on an electron in a particular state, takes it to the nearest neighbor state to the right. Is this operator Hermitian?

Enumerate the atoms from left to right as $i = 1, \dots, n$ and set $|i\rangle$ to be the state of an electron bound to atom i . Represented in this basis, the operator that shifts an electron to the nearest neighbor state to the right is given by

$$S_R = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix}$$

The operator is evidently non-hermitian as $(S_R)_{10} = 1 \neq 0^* = (S_R)_{01}$.

Exercise 8b Give an example of an operator that is both Hermitian and unitary. Show that an operator that is neither Hermitian nor unitary can always be written as a linear combination of a Hermitian and an anti-Hermitian operator

The matrix

$$\begin{pmatrix} 0 & e^{i\phi} \\ e^{-i\phi} & 0 \end{pmatrix}$$

is Hermitian and unitary for any angle ϕ .

Given an arbitrary operator A . Note that $A + A^\dagger$ is Hermitian and $A - A^\dagger$ is anti-Hermitian. Thus,

$$A = \frac{A + A^\dagger}{2} + \frac{A - A^\dagger}{2}$$

is the sum of a Hermitian and anti-Hermitian operator.

Exercise 8c Imagine you are rotating about an axis at a uniform angular velocity. Your friend, on the other hand, is stationary. Both of you observe a spin 1/2 particle. From your point of view the spin 1/2 particle is rotating about that axis in the opposite direction. After every 2π rotation, do you and your friend see different phase factors of the spin 1/2 particle?

Exercise 8d In a Stern-Gerlach experiment SG_z there are always some small magnetic field gradients along x and y directions due to $\nabla \cdot \vec{B} = 0$ constraint. How do these gradients effect the outcome of the SG_z device?

Remembering that $F = -\nabla(\mu \cdot \vec{B})$, we see that the x and y gradients lead to forces in these directions. These will lead to peaks in the x and y directions corresponding to these eigenstates.

Exercise 9a

Comparing components of the vector equation give the commutation relations exactly.

Exercise 9b Imagine I manage to create a state that is a simultaneous eigenstate of J_x and J_y . Determine the eigenvalues of this state.

Suppose $|\psi\rangle$ is such a state with $J_x|\psi\rangle = j_x|\psi\rangle, J_y|\psi\rangle = j_y|\psi\rangle$. We see then that

$$i\hbar J_z|\psi\rangle = [J_x, J_y]|\psi\rangle = 0|\psi\rangle$$

Thus, $|\psi\rangle$ must be in some combination of $|j, m = 0\rangle$ orbitals. We note that $j = 0$ gives eigenvalues of 0 for each J_i . Thus, $\psi = |j = 0, m = 0\rangle$.

Exercise 9c In a two dimensional Hilbert space I construct an operator that acting twice on the basis vectors annihilate them. On the other hand the anti-commutator of this operator with its adjoint behaves as an identity operator in this space. Can you determine the operator uniquely?

No you cannot. Let T be an operator with $T^2 = 0$ and $TT^\dagger + T^\dagger T = 1$. Consider the operator $\bar{T} = e^{i\theta}T$. Then, we see that $\bar{T}^2 = e^{i2\theta}T^2 = 0$ and $\{\bar{T}, \bar{T}^\dagger\} = e^{i\theta}Te^{-i\theta}T^\dagger + e^{-i\theta}T^\dagger e^{i\theta}T = TT^\dagger + T^\dagger T = 1$. So T is uniquely determined. Even if the Hilbert Space was over a real field, T is not uniquely determined. As $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ and $\frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$ both satisfy the desired properties.

Exercise 9d A Hermitian operator A is diagonalisable by a Unitary operator U . Show that if another operator B is diagonalisable by the same operator U then the operators A and B must commute.

Proof. By assumption, $A = UA_DU^\dagger$ and $B = UB_DU^\dagger$ where A_D and B_D are diagonal matrices. Note that A_D and B_D commute since they are diagonal. Then, we see that

$$AB = (UA_DU^\dagger)(UB_DU^\dagger) = UA_DB_DU^\dagger = UB_DA_DU^\dagger = BA$$

as desired. □